

Output devices

Devices used to produce results of the processed data are known as output devices. Some of the most commonly used output devices are.

- Monitor
- Printer
- Plotter
- Speaker
- Buzzer
- Motor
- Robot arm

Monitor

One of the most commonly used output device. Used to produce output on screen similar to the one used in television. It is also known as VDU (visual display unit)

Types of monitor

- CRT
- LCD (liquid crystal display)
- Plasma
- TFT (thin film transistor)
- LED (light emitting diode)

Difference between CRT and LCD / TFT / Plasma / LED

CRT (cathode ray tube)	LCD / TFT / Plasma / LED
Big in size	Small in Size
Inexpensive	Expensive
They emit radiation	Emit less radiation
Cause health problems • Headache • Eye strain	Cause health problems • Headache • Eye strain
Cannot be used where space is an issue	Used where space is an issue

Printer

It is another useful device used to produce output on a paper. The output that we get from printer is known as hardcopy

Types of printer

- Impact printer
- Non-impact printer

Impact printer

It uses a mechanism by which the print head strikes the paper while printing. They are noisy; the quality of printout is not good and is slower than non impact printers.

Types of impact printer

- Dot-matrix
- Ball printer
- Daisy wheel printer

Dot-matrix

- Fastest impact printer
- Can print up to 600 cps(characters per second)
- Noisy
- It uses pins to print. A dot-matrix printer may have 9, 18, 24 or 32 pins
- Dot matrix printer is used to print on multipart stationary and continuous stationary
- It produces high volume low quality output



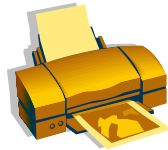
Ball printer and daisy wheel print head also strikes the paper they are also noisy and are slower than dot-matrix printer

Non-Impact printer

In non-impact printers the print head does not strike the paper to print. Instead of pins they use spray nozzles. They are less noisy, are faster than impact printers and the print we get from these printers is of good quality.

Types of non-impact printer

- LASER printer
 - Inkjet printer
- (They are used to produce high quality but low volume output. Should be used carefully as ink can smudge, they are slower as compared to laser printer)



LASER printer

Fastest non-impact printer can print 20000 lines per minute, less noisy and the printout we get from this printer is of good quality. They are used to print high volume, high quality printouts



Plotters

It is used to print large size, high quality printout. They are used with CAD (Computer aided design) software to print graphs, maps, building design etc. there are two types



Flat-bed and drum plotter

Buzzer

Used in washing machines and microwave oven to sound an alarm.

Motors:

They convert electrical signals to motion signals. Used in greenhouse to open and close windows, used in washing machines and microwave ovens.

Robot Arm:

It is an electro mechanical device. Used in industries to perform various processes/tasks. A robot arm has 7 parts with six joints and at each joint a motor is installed. A robot arm is used to perform the following tasks.

- Lifting heavy objects
- Drilling holes
- Painting car parts
- Tightening bolts etc